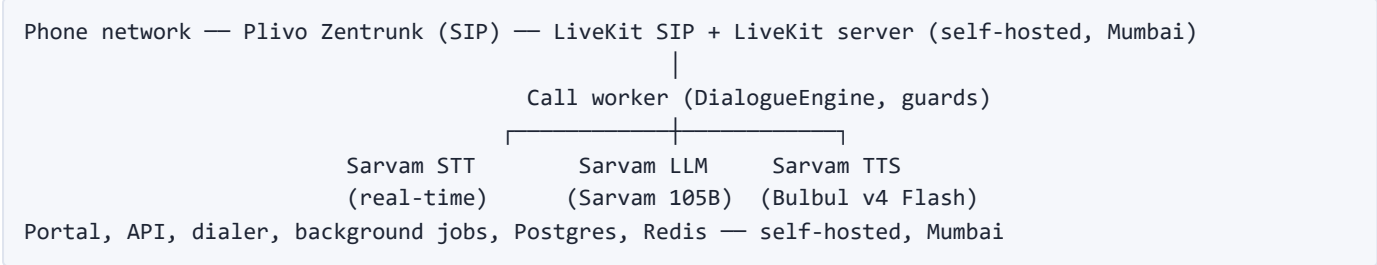
Sarvam + Plivo Cost Model

Hinglish AI sales calling agent · monthly running cost at 0, 1, 2, 10 and 20 clients · NavyaNexus AI · 11 November 2026

Architecture priced here



Summary

Clients	Lines (calls at once)	VMs	Answered minutes / month	Usage cost	Fixed cost	Total per month	In US dollars	Per client
0	0	1	0	₹0	₹3,300 – ₹11,350	**₹3,300 – ₹11,350**	$38 – $129	—
1	20	3–4	1,74,720	₹3.42 lakh – ₹3.83 lakh	₹26,350 – ₹47,600	**₹3.69 lakh – ₹4.30 lakh**	$4,191 – $4,889	₹3.69 lakh – ₹4.30 lakh
2	40	4–6	3,49,440	₹6.85 lakh – ₹7.65 lakh	₹40,600 – ₹66,100	**₹7.26 lakh – ₹8.31 lakh**	$8,244 – $9,447	₹3.63 lakh – ₹4.16 lakh
10	200	17	17,47,200	₹34.25 lakh – ₹38.26 lakh	₹1,27,600 – ₹2,30,100	**₹35.52 lakh – ₹40.56 lakh**	$40,365 – $46,096	₹3.55 lakh – ₹4.06 lakh
20	400	30–31	34,94,400	₹68.49 lakh – ₹76.53 lakh	₹2,30,100 – ₹3,95,100	**₹70.79 lakh – ₹80.48 lakh**	$80,445 – $91,453	₹3.54 lakh – ₹4.02 lakh

Low = Sarvam Business plan rates and low server estimates. High = Sarvam pay-as-you-go list prices and high server estimates. Each client runs 20 calls at once, 8 hours a day, 26 days a month.

Key points

- Usage cost is **₹1.96 – ₹2.19 per answered minute**, plus servers.
- Each client at full use (20 lines, 8 hours) costs about **₹3.54 lakh – ₹4.30 lakh per month**.
- Usage cost per minute splits as Plivo phone line 36%, Sarvam TTS 35%, Sarvam STT 19%, Sarvam LLM 10%.
- At ₹3.00 – 3.50 per answered minute charged to clients, the margin is roughly **18 – 42%**.
- With no clients the system costs **₹3,300 – ₹11,350 per month** to keep online.

Assumptions

Call volume

Assumption	Value	Note
Calls at once per client	20	Each client runs 20 lines in parallel during calling hours
Calling hours per day	8	Inside the legal 9 am – 9 pm window
Calling days per month	26	Monday to Saturday
Share of line time in answered talk	70%	The rest is dialing, ringing, no-answers and gaps
Answered minutes per line per month	8,736	8 × 60 × 0.7 × 26
Answered minutes per client per month	1,74,720	20 lines
Exchange rate	₹88 = $1	For the dollar columns

Usage cost per answered minute

Item	How it is worked out	Business plan ₹/min	Pay-as-you-go ₹/min
Plivo phone line	₹0.60/min outbound India, billed in 60-second pulses; rounding up adds ~17% on 2-3 minute calls	₹0.70	₹0.70
Sarvam TTS (Bulbul v4 Flash)	~400 live characters per minute × ₹1.70 (Business) – ₹2.00 (pay-as-you-go) per 1,000 characters	₹0.68	₹0.80
Sarvam STT (real-time)	₹0.38 (Business) – ₹0.49 (pay-as-you-go) per minute, charged for the whole call	₹0.38	₹0.49
Sarvam LLM (Sarvam 105B)	~16k input tokens/min (60% cached) + ~600 output tokens/min at ₹15 / ₹5 / ₹60 per 1M	₹0.20	₹0.20
Usage cost per answered minute		**₹1.96**	**₹2.19**

How the LLM cost is worked out

LLM token type	Volume	Price	₹ per minute
Uncached input	6,400 tokens/min	₹15 per 1M	₹0.096
Cached input	9,600 tokens/min	₹5 per 1M	₹0.048
Output	600 tokens/min	₹60 per 1M	₹0.036
Total			**₹0.18 (rounded to ₹0.20)**

About 3.5 model calls per minute (replies and tool rounds) plus the observer, with ~3,000 tokens of prompt per call. If Sarvam 105B uses reasoning tokens, the LLM line can rise to ~₹0.40 per minute.

How the TTS characters are worked out

TTS step	Value
Agent speaking time	~45% of each call minute
Speaking speed (Bulbul at pace 1.0)	~18 characters per second
Characters spoken per call minute	~490
Covered by cached phrases (greeting, disclosure, scripted lines)	~20%, not charged again
Live characters charged per call minute	**~400**

Where the usage cost goes

Item	₹ per minute (Business)	Share of usage cost
Plivo phone line	₹0.70	36%
Sarvam TTS (Bulbul v4 Flash)	₹0.68	35%
Sarvam STT (real-time)	₹0.38	19%
Sarvam LLM (Sarvam 105B)	₹0.20	10%

0 clients

Nothing is being called, so every per-use charge is zero. What remains is the cost of keeping the system online for demos and testing.

TOTAL PER MONTH **₹3,300 – ₹11,350** $38 – $129

Fixed costs

Item	VMs	Detail	₹ per month
Server	**1** × 2–4 vCPU, 8 GB	One small Mumbai server (2–4 cores, 8 GB) running API, worker, background jobs, Postgres and Redis	₹3,000 – ₹7,000
Managed Postgres	— (service)	Optional, instead of the database on the server	₹0 – ₹4,000
LiveKit	— (service)	LiveKit Cloud free Build plan (5 calls at once, 1,000 SIP minutes); worker stays on your server	₹0
Plivo number	—	One number for demos and test calls	₹200 – ₹250
Domain	—	Domain name (SSL certificate is free)	₹100
Monitoring	—	Free tiers	₹0
Total	**1 VM**		**₹3,300 – ₹11,350**

Usage costs

Item	With no calls
Sarvam STT, TTS, LLM	₹0 — charged per use, pay-as-you-go has no monthly minimum
Plivo call minutes	₹0 — charged per answered minute
LiveKit	₹0 — free Build plan, or ₹3,000–5,000 more for a self-hosted server with a public IP

Costs that can appear even with no clients

Cost that can still appear	How much	How to control it
Your own test calls on a phone	~₹2.19 per minute	Keep test calls short
Your own test calls in the browser (no Plivo)	~₹1.26 – ₹1.49 per minute	—
Landing-page “Talk to Priya” demo	~₹1.26 – ₹1.49 per visitor minute	DEMO_MAX_CONCURRENT, DEMO_CALLS_PER_HOUR, or DEMO_CALL_ENABLED=false
Modal GPU left running (T4, 24/7)	~₹35,000 – 40,000 per month	Not needed with Sarvam; leave NAVYA_MIN_CONTAINERS unset
Sarvam Growth or Business plan taken too early	₹49,999 or ₹1,99,999 minimum per month	Stay on pay-as-you-go until there is volume
Full 300–400 call setup left running idle	₹2.5 – 4 lakh per month	Start on one server; add workers as clients join

1 client

TOTAL PER MONTH **₹3.69 lakh – ₹4.30 lakh** $4,191 – $4,889

Scenario

Item	Value
Clients	1
Lines (calls at once)	20 (1 × 20)
VMs in total	**3–4**
Answered minutes per month	1,74,720
Sarvam plan	Business (₹1,99,999/month minimum, returned as credits)
Sarvam spend per month (STT + TTS + LLM)	₹2,20,147 – ₹2,60,333

Usage costs (1,74,720 answered minutes)

Item	₹ per minute	₹ per month
Plivo phone line	₹0.70	₹1,22,304
Sarvam TTS (Bulbul v4 Flash)	₹0.68	₹1,18,810 – ₹1,39,776
Sarvam STT (real-time)	₹0.38	₹66,394 – ₹85,613
Sarvam LLM (Sarvam 105B)	₹0.20	₹34,944
Usage total	**₹1.96 – ₹2.19**	**₹3,42,451 – ₹3,82,637**

Fixed costs for 20 calls at once

Item	VMs	Detail	₹ per month
Call workers	**1–2** × 4 vCPU, 8 GB	1–2 servers with 4 cores each (~20 calls per server, one spare)	₹10,000 – ₹20,000
LiveKit server + SIP	**1** × 4 vCPU, 8 GB, public IP	One 4-core server with a public IP, self-hosted (software is free)	₹8,000 – ₹12,000
API, background, database	**1** × 4 vCPU, 16 GB (Postgres and Redis on it)	API, background jobs, Postgres, Redis, storage, monitoring	₹8,000 – ₹15,000
Plivo numbers	—	1–2 numbers at ₹200–250 each	₹250 – ₹500
Domain	—	Domain name	₹100
Fixed total	**3–4 VMs**		**₹26,350 – ₹47,600**

Total

	Low (Business rates, low fixed)	High (pay-as-you-go, high fixed)
Usage	₹3,42,451	₹3,82,637
Fixed	₹26,350	₹47,600
Total per month	**₹3,68,801**	**₹4,30,237**
In lakh	₹3.69 lakh	₹4.30 lakh
In US dollars	$4,191	$4,889
Per client	₹3,68,801	₹4,30,237
All-in cost per answered minute	₹2.11	₹2.46

If clients call fewer hours

Calling hours per day	Answered minutes	Total per month	Per client
8 hours	1,74,720	₹3.69 lakh – ₹4.30 lakh	₹3.69 lakh – ₹4.30 lakh
4 hours	87,360	₹1.98 lakh – ₹2.39 lakh	₹1.98 lakh – ₹2.39 lakh
2 hours	43,680	₹1.12 lakh – ₹1.43 lakh	₹1.12 lakh – ₹1.43 lakh

Fixed costs are kept the same in this table; with fewer hours, workers can also be shut down sooner.

What you would make

Price charged to clients	Revenue per month	Margin per month	Margin %
₹3.00 per answered minute	₹5.24 lakh	₹0.94 lakh – ₹1.55 lakh	18% – 30%
₹3.50 per answered minute	₹6.12 lakh	₹1.81 lakh – ₹2.43 lakh	30% – 40%

2 clients

TOTAL PER MONTH **₹7.26 lakh – ₹8.31 lakh** $8,244 – $9,447

Scenario

Item	Value
Clients	2
Lines (calls at once)	40 (2 × 20)
VMs in total	**4–6**
Answered minutes per month	3,49,440
Sarvam plan	Business (₹1,99,999/month minimum, returned as credits)
Sarvam spend per month (STT + TTS + LLM)	₹4,40,294 – ₹5,20,666

Usage costs (3,49,440 answered minutes)

Item	₹ per minute	₹ per month
Plivo phone line	₹0.70	₹2,44,608
Sarvam TTS (Bulbul v4 Flash)	₹0.68	₹2,37,619 – ₹2,79,552
Sarvam STT (real-time)	₹0.38	₹1,32,787 – ₹1,71,226
Sarvam LLM (Sarvam 105B)	₹0.20	₹69,888
Usage total	**₹1.96 – ₹2.19**	**₹6,84,902 – ₹7,65,274**

Fixed costs for 40 calls at once

Item	VMs	Detail	₹ per month
Call workers	**2–3** × 4 vCPU, 8 GB	2–3 servers with 4 cores each	₹20,000 – ₹30,000
LiveKit server + SIP	**1** × 4–8 vCPU, 16 GB, public IP	One 4–8 core server with a public IP, self-hosted	₹10,000 – ₹15,000
API, background, database	**1–2** × 4 vCPU, 16 GB (Postgres and Redis on it)	API, background jobs, Postgres, Redis, storage, monitoring	₹10,000 – ₹20,000
Plivo numbers	—	2–4 numbers (one or two per client)	₹500 – ₹1,000
Domain	—	Domain name	₹100
Fixed total	**4–6 VMs**		**₹40,600 – ₹66,100**

Total

	Low (Business rates, low fixed)	High (pay-as-you-go, high fixed)
Usage	₹6,84,902	₹7,65,274
Fixed	₹40,600	₹66,100
Total per month	**₹7,25,502**	**₹8,31,374**
In lakh	₹7.26 lakh	₹8.31 lakh
In US dollars	$8,244	$9,447
Per client	₹3,62,751	₹4,15,687
All-in cost per answered minute	₹2.08	₹2.38

If clients call fewer hours

Calling hours per day	Answered minutes	Total per month	Per client
8 hours	3,49,440	₹7.26 lakh – ₹8.31 lakh	₹3.63 lakh – ₹4.16 lakh
4 hours	1,74,720	₹3.83 lakh – ₹4.49 lakh	₹1.92 lakh – ₹2.24 lakh
2 hours	87,360	₹2.12 lakh – ₹2.57 lakh	₹1.06 lakh – ₹1.29 lakh

Fixed costs are kept the same in this table; with fewer hours, workers can also be shut down sooner.

What you would make

Price charged to clients	Revenue per month	Margin per month	Margin %
₹3.00 per answered minute	₹10.48 lakh	₹2.17 lakh – ₹3.23 lakh	21% – 31%
₹3.50 per answered minute	₹12.23 lakh	₹3.92 lakh – ₹4.98 lakh	32% – 41%

10 clients

TOTAL PER MONTH **₹35.52 lakh – ₹40.56 lakh** $40,365 – $46,096

Scenario

Item	Value
Clients	10
Lines (calls at once)	200 (10 × 20)
VMs in total	**17**
Answered minutes per month	17,47,200
Sarvam plan	Business (concurrency limits to confirm with Sarvam)
Sarvam spend per month (STT + TTS + LLM)	₹22,01,472 – ₹26,03,328

Usage costs (17,47,200 answered minutes)

Item	₹ per minute	₹ per month
Plivo phone line	₹0.70	₹12,23,040
Sarvam TTS (Bulbul v4 Flash)	₹0.68	₹11,88,096 – ₹13,97,760
Sarvam STT (real-time)	₹0.38	₹6,63,936 – ₹8,56,128
Sarvam LLM (Sarvam 105B)	₹0.20	₹3,49,440
Usage total	**₹1.96 – ₹2.19**	**₹34,24,512 – ₹38,26,368**

Fixed costs for 200 calls at once

Item	VMs	Detail	₹ per month
Call workers	**13** × 4 vCPU, 8 GB (fewer outside calling hours)	10 servers needed + 25% headroom = ~13 servers with 4 cores; lower end if they shut down outside calling hours	₹65,000 – ₹1,30,000
LiveKit server + SIP	**2** × 8 vCPU, 16 GB, public IP	Two 8-core servers with public IPs, self-hosted	₹25,000 – ₹35,000
API, background, database	**2** × 4 vCPU, 8 GB app servers; Postgres and Redis managed	Managed Postgres, Redis, API, background jobs, storage, monitoring	₹35,000 – ₹60,000
Plivo numbers	—	10–20 numbers	₹2,500 – ₹5,000
Domain	—	Domain name	₹100
Fixed total	**17 VMs**		**₹1,27,600 – ₹2,30,100**

Total

	Low (Business rates, low fixed)	High (pay-as-you-go, high fixed)
Usage	₹34,24,512	₹38,26,368
Fixed	₹1,27,600	₹2,30,100
Total per month	**₹35,52,112**	**₹40,56,468**
In lakh	₹35.52 lakh	₹40.56 lakh
In US dollars	$40,365	$46,096
Per client	₹3,55,211	₹4,05,647
All-in cost per answered minute	₹2.03	₹2.32

If clients call fewer hours

Calling hours per day	Answered minutes	Total per month	Per client
8 hours	17,47,200	₹35.52 lakh – ₹40.56 lakh	₹3.55 lakh – ₹4.06 lakh
4 hours	8,73,600	₹18.40 lakh – ₹21.43 lakh	₹1.84 lakh – ₹2.14 lakh
2 hours	4,36,800	₹9.84 lakh – ₹11.87 lakh	₹0.98 lakh – ₹1.19 lakh

Fixed costs are kept the same in this table; with fewer hours, workers can also be shut down sooner.

What you would make

Price charged to clients	Revenue per month	Margin per month	Margin %
₹3.00 per answered minute	₹52.42 lakh	₹11.85 lakh – ₹16.89 lakh	23% – 32%
₹3.50 per answered minute	₹61.15 lakh	₹20.59 lakh – ₹25.63 lakh	34% – 42%

20 clients

TOTAL PER MONTH **₹70.79 lakh – ₹80.48 lakh** $80,445 – $91,453

Scenario

Item	Value
Clients	20
Lines (calls at once)	400 (20 × 20)
VMs in total	**30–31**
Answered minutes per month	34,94,400
Sarvam plan	Enterprise (needed for 400 streams at once; rates assumed equal to Business)
Sarvam spend per month (STT + TTS + LLM)	₹44,02,944 – ₹52,06,656

Usage costs (34,94,400 answered minutes)

Item	₹ per minute	₹ per month
Plivo phone line	₹0.70	₹24,46,080
Sarvam TTS (Bulbul v4 Flash)	₹0.68	₹23,76,192 – ₹27,95,520
Sarvam STT (real-time)	₹0.38	₹13,27,872 – ₹17,12,256
Sarvam LLM (Sarvam 105B)	₹0.20	₹6,98,880
Usage total	**₹1.96 – ₹2.19**	**₹68,49,024 – ₹76,52,736**

Fixed costs for 400 calls at once

Item	VMs	Detail	₹ per month
Call workers	**25** × 4 vCPU, 8 GB (fewer outside calling hours)	20 servers needed + 25% headroom = ~25 servers with 4 cores; lower end if they shut down outside calling hours	₹1,20,000 – ₹2,40,000
LiveKit server + SIP	**3** × 8 vCPU, 16 GB, public IP	Three 8-core servers with public IPs, self-hosted	₹45,000 – ₹55,000
API, background, database	**2–3** × 4 vCPU, 8 GB app servers; Postgres and Redis managed	Managed Postgres, Redis, API, background jobs, storage, monitoring	₹60,000 – ₹90,000
Plivo numbers	—	20–40 numbers	₹5,000 – ₹10,000
Domain	—	Domain name	₹100
Fixed total	**30–31 VMs**		**₹2,30,100 – ₹3,95,100**

Total

	Low (Business rates, low fixed)	High (pay-as-you-go, high fixed)
Usage	₹68,49,024	₹76,52,736
Fixed	₹2,30,100	₹3,95,100
Total per month	**₹70,79,124**	**₹80,47,836**
In lakh	₹70.79 lakh	₹80.48 lakh
In US dollars	$80,445	$91,453
Per client	₹3,53,956	₹4,02,392
All-in cost per answered minute	₹2.03	₹2.30

If clients call fewer hours

Calling hours per day	Answered minutes	Total per month	Per client
8 hours	34,94,400	₹70.79 lakh – ₹80.48 lakh	₹3.54 lakh – ₹4.02 lakh
4 hours	17,47,200	₹36.55 lakh – ₹42.21 lakh	₹1.83 lakh – ₹2.11 lakh
2 hours	8,73,600	₹19.42 lakh – ₹23.08 lakh	₹0.97 lakh – ₹1.15 lakh

Fixed costs are kept the same in this table; with fewer hours, workers can also be shut down sooner.

What you would make

Price charged to clients	Revenue per month	Margin per month	Margin %
₹3.00 per answered minute	₹104.83 lakh	₹24.35 lakh – ₹34.04 lakh	23% – 32%
₹3.50 per answered minute	₹122.30 lakh	₹41.83 lakh – ₹51.51 lakh	34% – 42%

LiveKit: self-hosted vs LiveKit Cloud

Why LiveKit appears in the costs

Question	Answer
LiveKit software (server + SIP)	Free, open source (Apache 2.0); no licence or per-minute fee when self-hosted
LiveKit lines in this document	Rent for the cloud server that runs the free software
LiveKit Cloud	LiveKit's hosted service, charged per minute; only if you don't self-host
Why not a laptop	Phone audio needs a public IP and open UDP ports, the machine must stay on all day, and a laptop handles only a few calls
When own hardware is enough	An office machine with a static public IP, open ports, reliable power and enough CPU costs only electricity and internet

What LiveKit Cloud would add instead of self-hosting

Clients	Calls at once	LiveKit Cloud plan needed	Extra cost per month vs self-hosting
1	20	Ship ($50, max 20 calls at once)	₹2.04 lakh
2	40	Scale ($500, up to 600 calls at once)	₹4.44 lakh
10	200	Scale ($500, up to 600 calls at once)	₹20.43 lakh
20	400	Scale ($500, up to 600 calls at once)	₹40.42 lakh

Agent session $0.01/min + third-party SIP $0.003/min ≈ ₹1.14 per minute, plus the plan fee. The free Build plan (5 calls at once) covers testing at 0 clients. The worker uses local voice detection, so no paid LiveKit models are used.

Ways to lower the cost later

Option	What it does	Saving
Per-second billing from Plivo	Removes the 60-second rounding	~₹0.10 per minute
Send audio to STT only while the customer speaks	If Sarvam charges only for open stream time (not confirmed)	Up to ~₹0.20 per minute
Cache more common lines	Fewer live TTS characters	Small, grows with script reuse
Sarvam volume pricing	Enterprise rates at lakhs of minutes per month	Negotiated
Shut down workers outside calling hours	Servers run ~12 hours instead of 24	Up to half of worker cost

Check before relying on these numbers

Check before relying on these numbers	Why
Bulbul v4 Flash discount	50% off until 12 Nov 2026; this document uses the full price
Sarvam concurrency limits	Not published per plan; 200–400 STT and TTS streams at once likely need Enterprise
Sarvam 105B in live calls	Check its response time, and whether reasoning tokens raise the output cost (LLM up to ₹0.40/min)
Plivo rate	₹0.60/min on Plivo's India page, an older page shows ₹0.40; confirm in the console
Rates in config/settings.py	Out of date (Plivo ₹0.45, LLM ₹4/₹16 per 1M); cost reports from the code show too little until updated
Server prices	Estimates for Mumbai cloud servers on demand; reserved or committed pricing is lower

Sources

Sarvam API pricing — sarvam.ai/api-pricing

Plivo SIP trunking pricing, India — plivo.com/sip-trunking/pricing/in

Plivo voice pricing, India — plivo.com/voice/pricing/in

LiveKit Cloud pricing — livekit.io/pricing

Calls per server, cached phrases and call flow — this project's settings and cost report

Prices are as listed on 11 November 2026, excluding GST.